

Run report

2026-10-04T024528Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	2	dataset	gaussian
seed	223,556,844	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	7.629×10^{-6}	best_epoch	60
compile_cache_warm	True		
gaussian_params	dim=2, mu_gen=[0, 1], mu_true=[0.2, 0.8], cov_gen=[[1, -0.9], [-0.9, 2.25]], cov_true=[[0.81, -0.702], [-0.702, 1.69]], cov_detector=[[0.25, 0], [0, 0.64]]		
mmd_sigmas_detector	$1.460 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$1.217 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
<i>RAN</i>			
train	14.686	68.3	train
transfer	0.531		train
compile	2.852		train
epochs	6.491		train
select	4.118		train
particle_mmd	0.489	2.3	train
save	0.128	0.6	train
data	0.121	0.6	gaussian cache hit – load
load	1.369	6.4	load
plots	4.716	21.9	load
evaluate	0.003	0.0	load
total	21.511	100.0	
<i>Omnifold baseline</i>			
parse_config	0.002	0.0	omnifold
data	0.148	0.0	gaussian cache hit – omnifold
omnifold	439.156	98.4	niter=3, epochs=50 – omnifold
init	1.250		DataLoaders and the two MLPs – omnifold
unfold	419.916		MultiFold.Unfold – omnifold
iter1_step1	60.954		detector-level reweighting – omnifold
iter2_step1	59.559		detector-level reweighting – omnifold
iter3_step1	65.063		detector-level reweighting – omnifold
iter1_step2	52.103		particle-level reweighting – omnifold
iter2_step2	65.703		particle-level reweighting – omnifold
iter3_step2	116.503		particle-level reweighting – omnifold
reweight	1.587		evaluating weights on z_target – omnifold
evaluate	7.006	1.6	2 variables – omnifold
total	446.312	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	207.1	7.804	96.2	5.483	97.4	14.08	93.2
dim_1	230.1	27.45	88.1	10.78	95.3	16.73	92.7

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	6.676	0.2937	95.6	0.2560	96.2	0.3099	95.4
dim_1	5.198	0.6584	87.3	0.2631	94.9	0.3365	93.5

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	26.17	1.150	95.6	1.003	96.2	1.214	95.4
dim_1	20.22	2.506	87.6	1.011	95.0	1.291	93.6

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	162.5	61.78	62.0	6.920	95.7	12.77	92.1

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	204.8	20.17	90.2	8.418	95.9	12.57	93.9
dim_1	242.0	48.17	80.1	12.44	94.9	20.87	91.4

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	8.734	1.930	77.9	0.2204	97.5	0.2946	96.6
dim_1	8.100	2.746	66.1	0.2226	97.3	0.3602	95.6

VLC divergence [$\times 10^3$]

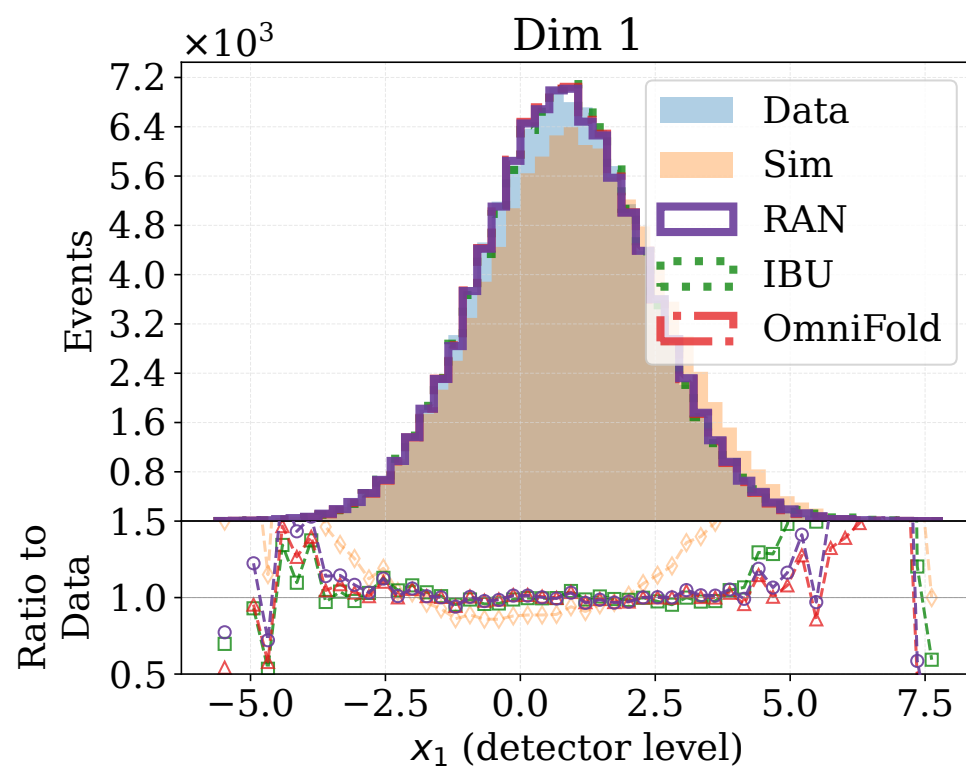
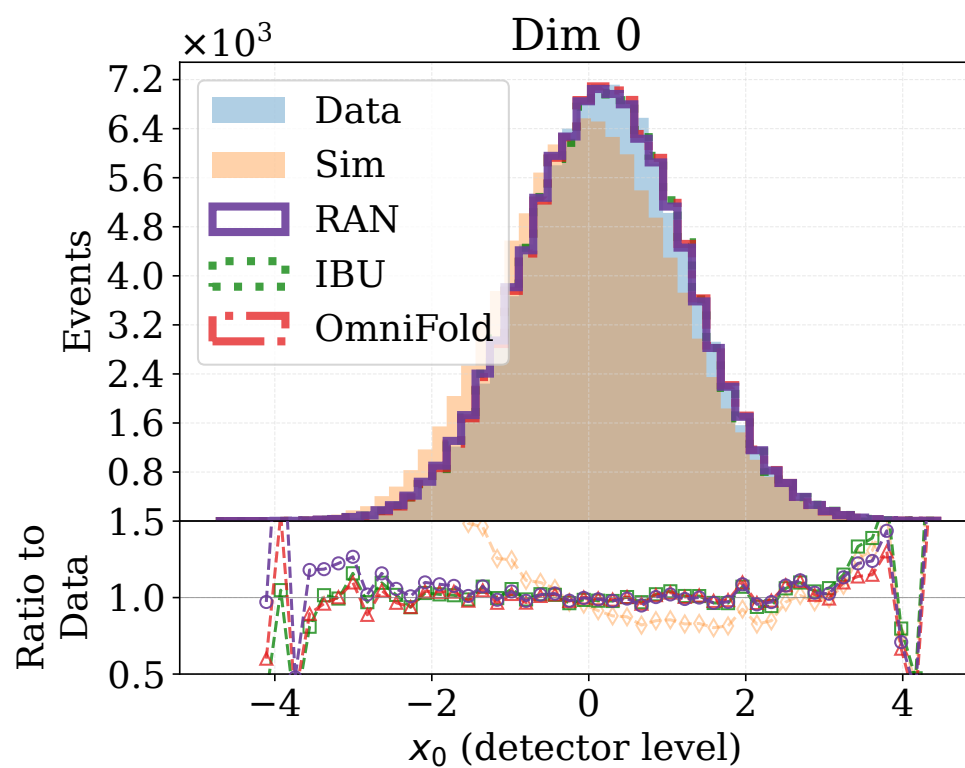
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	34.05	7.635	77.6	0.8533	97.5	1.135	96.7
dim_1	31.35	10.78	65.6	0.8743	97.2	1.407	95.5

Sliced Wasserstein, joint [$\times 10^3$]

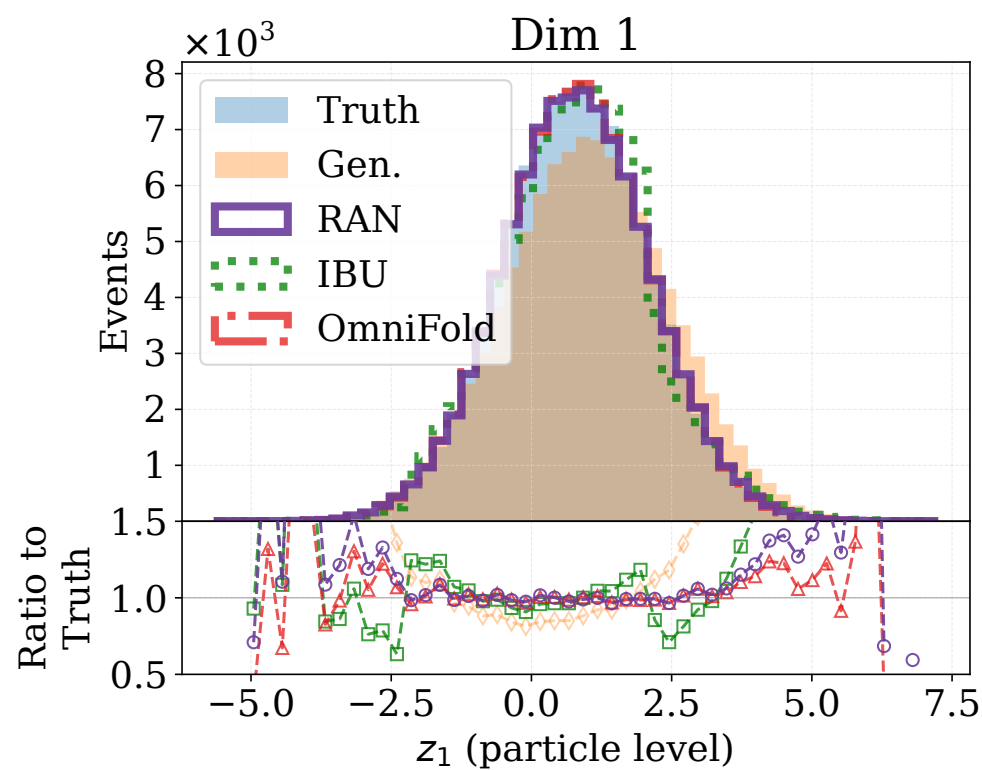
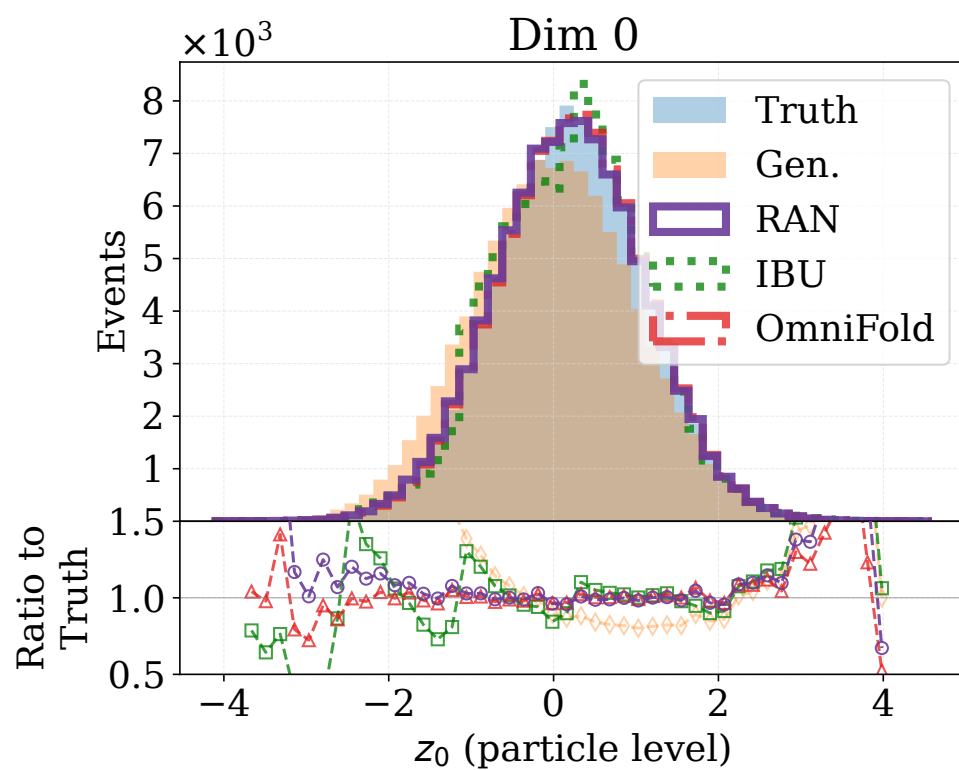
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	193.1	76.13	60.6	11.09	94.3	16.03	91.7

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

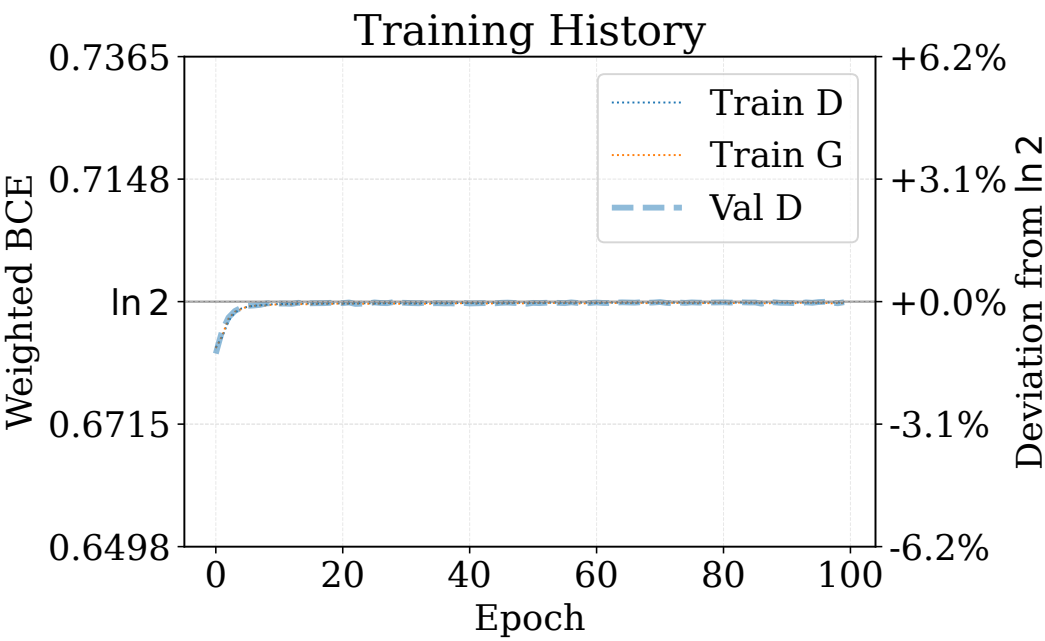
Detector Level



Particle Level



Training diagnostics
Training losses



Selection criteria

